LUFUNO RODGER MPHAGI

BSc Computer Science Finalist | Software Engineer | Backend, AI & Full-Stack

Pretoria, Gauteng, South Africa | lufunomphagi@gmail.com | linkedin.com/in/LufunoMphagi | github.com/LufunoMphagi

Profile

BSc Computer Science finalist at the University of Pretoria with hands-on experience in software engineering, backend development, database design, AI-enabled features, CI/CD, full-stack web development, production support, and technical communication. Backend developer and AI subteam lead on a year-long COS301 capstone, contributing to a mobile budgeting platform built with Flutter, Dart, Drift, SQLite, GitHub Actions, embeddings, and on-device AI concepts. Experienced across the software development lifecycle: requirements, architecture, implementation, testing, debugging, code review, integration, deployment, documentation, and stakeholder support. Strong problem solver with practical experience in Java, Python, C++, JavaScript, SQL, React, Next.js, Flutter, Firebase, REST/JSON, Git, and relational data systems.

Technical Skills

Programming	Java, Python, C++, JavaScript, Dart, SQL, PHP, TypeScript basics, C#/.NET foundations
Backend & APIs	REST/JSON concepts, service-layer logic, validation, authentication concepts, data-access patterns, client/server systems, API design fundamentals
Databases & Data	SQLite, Drift, Firebase/Firestore, SQL, relational modelling, schema design, CRUD, joins/queries, data validation, persistence, reporting
AI & Intelligent Systems	Embeddings, vector similarity, transaction classification, inference pipelines, feature extraction, caching, AI-assisted analytics, prompt engineering basics
Frontend & Mobile	Flutter, React, Next.js, HTML5, CSS3, Tailwind CSS, responsive UI, reusable components, accessibility/usability basics
Software Engineering	OOP, algorithms, data structures, SDLC, Agile teamwork, design patterns, debugging, unit/widget testing, code review, documentation
DevOps & Tooling	Git, GitHub, branching, pull requests, GitHub Actions, CI/CD concepts, VS Code, Postman/Thunder Client, Linux/WSL, Firebase, Netlify/Vercel
Networks & Systems	TCP/IP, DNS, HTTP/HTTPS, SMTP, LDAP, Apache, Wireshark, Windows support, Linux fundamentals, client/server troubleshooting

Flagship Software Engineering Experience

Backend Developer & AI Subteam Lead - Mobile Budgeting Application 2026
COS301 Software Engineering Capstone, University of Pretoria Flutter, Dart, Drift, SQLite, AI

- Contributed to a year-long team software engineering project for personal finance management, including budgeting, transaction tracking, reporting, categorisation, and AI-assisted insights.
- Designed and implemented database-backed functionality using Drift/SQLite, including categories, hierarchical category relationships, transactions, transaction-category mappings, budget templates/periods, application settings, and embedding-cache data.
- Built and integrated real-data workflows for budget limits, category spending, transaction actions, financial reporting, and persistence, replacing mock data with production-style application logic.
- Implemented and tested backend/service logic for spending features, predictions, statement interpretation, duplicate detection, transaction inference, and import-preview orchestration.
- Helped develop an AI transaction-classification pipeline using text embeddings, vector similarity, category scoring, embedding-text construction, caching, and on-device model integration concepts.
- Worked with a BGE-small English embedding model asset and ONNX-based integration, with tests covering embedding-text construction and vector-similarity behaviour.
- Used Git feature branches, pull requests, merge-conflict resolution, code review, automated testing, build tooling, CI/CD, and technical documentation throughout collaborative delivery.
- Took ownership of backend integration and later coordinated a three-person AI subteam, balancing implementation, testing, debugging, integration, and demo readiness.

Professional Experience

Cambridge International Computer Science Teacher Sep 2026 – Present
The British Academy Pretoria

- Teach programming, algorithms, data representation, software development, arrays, debugging, defensive programming, searching/sorting, and problem-solving.
- Create technical notes, practical coding exercises, assessments, and demonstrations, strengthening communication and the ability to explain complex concepts clearly.

Assistant Webmaster Feb 2026 – Present
University of Pretoria Pretoria

- Support live university web systems used by students, staff, departments, and public visitors; troubleshoot publishing issues, broken links, layout defects, content problems, and user-facing inconsistencies.
- Translate stakeholder requests into technical tasks, test changes, document outcomes, and communicate clearly with non-technical users.
- Improve usability, responsive behaviour, content quality, accessibility basics, and reliability across production-facing digital services.

Freelance Web & Software Developer 2025 – Present
Mphagi Technologies / Client Projects South Africa

- Build and maintain client-facing websites and web applications using React, Next.js, JavaScript, Firebase, Tailwind CSS, databases, authentication concepts, hosting, deployment, and integrations.
- Deliver projects from requirements gathering and implementation through debugging, deployment, client feedback, documentation, and post-launch support.

Web Content Specialist Sep 2025 – Dec 2025
University of Pretoria Pretoria

- Implemented and quality-checked website updates across multiple university pages, improving navigation, responsive behaviour, content accuracy, and visual consistency.

Student IT Lab Technician Jul 2023 – Sep 2025
University of Pretoria Pretoria

- Provided first-line support for software, login, browser, device, lab, and connectivity issues; reproduced problems, tested fixes, documented outcomes, and escalated when required.

Selected Projects

House of Makie 2025 – 2026
E-commerce Web Application Next.js, Firebase, JavaScript

- Built responsive product experiences, authentication and admin-oriented functionality, Firebase-backed content, deployment workflows, and customer-facing features.

Workshop Collective Web Platform 2025 – 2026
Full-Stack Client Platform React, Tailwind CSS, Firebase

- Developed reusable React components, responsive interfaces, database-driven content, admin features, real-time sync concepts, deployment preparation, and client-requested enhancements.

SM Studio Digital Platform 2026
Business Website, Booking & Client Workflows Next.js, Tailwind CSS, Firebase

- Built service, gallery, contact and booking-oriented flows; worked on hosting/domain configuration, analytics/dashboard concepts, quote/payment planning, deployment troubleshooting, and client portal functionality.

FieldOps Work Order System 2026
Software Engineering Project API Design, Data Modelling, Team Development

- Contributed to work-order, task-allocation, and operational workflow design, including validation, role-based behaviour, API concepts, requirements analysis, testing, and technical documentation.

Education

BSc Computer Science - Finalist 2022 – 2026
University of Pretoria Pretoria
Relevant coursework: Software Engineering, Databases, Data Structures & Algorithms, Artificial Intelligence, Computer Networks, Programming Languages, Systems Design, and Operating Systems concepts.

Certifications & Leadership

Microsoft 365 Fundamentals (2024) | Career Essentials in Data Analysis - Microsoft | Career Essentials in System Administration - Microsoft
Artificial Intelligence in the 4IR - University of Johannesburg | C++ Programming - Cisco
Chairperson, School of IT Sub-House (2023–2024) | Top 3 Mentor, EBIT Faculty (2024) | Finalist, WITS Mathematics Competition (2022)